

A 9-State Markov Chain Model for Semi-Fixed Menstruation Patterns: Stationary Distribution, Mean First Passage Times, and Population Heterogeneity

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Abstract

We introduce and analyse a nine-state, discrete-time, homogeneous Markov chain that models the longitudinal dynamics of *semi-fixed menstruation* (*veset hatzi kavu'a*), a central construct of Jewish Halachic law governing anticipatory pre-menstrual abstinence. The state space encodes both the ordinal position of a cycle within a semi-fixed stretch and the sub-type of the accumulated cycle-length range: short ($r_{\max} < 29$), normal ($29 \leq r_{\min}$ or $r_{\max} \leq 31$), or long ($r_{\min} > 31$), yielding nine states whose transition structure is fully determined by the Halachic definition— independent of data.

We prove that the 9×9 transition matrix has an exact 3×3 block structure that reduces the stationary computation to a 3×3 eigenproblem, provide a closed-form block formula for the stationary distribution π , and establish irreducibility and aperiodicity. Using a Natural Family Planning (NFP) dataset of 1,221 cycles from 98 participants (after excluding *Haflaga*-established clients), we estimate $\hat{\mathbf{P}}$ and obtain: $\pi_{S_{32}} \approx 54.2\%$ (normal semi-fixed, full Halachic obligations), $\pi_{S_{31}} + \pi_{S_{33}} \approx 8.9\%$ (favourable semi-fixed, no extra obligations), with 95% bootstrap confidence intervals of $[48.8\%, 59.1\%]$ and $[6.1\%, 12.1\%]$, respectively. The spectral gap $\delta = 1 - |\lambda_2| \approx 0.297$ implies the chain reaches stationarity within ≤ 23 cycles, validated by a Chapman-Kolmogorov test ($p = 0.97$, confirming the order-1 Markov assumption).

We compute mean first passage times (MFPTs) to the favourable set $\{S_{31}, S_{33}\}$ from every state: a woman starting fresh (state S_{11}) expects to reach a favourable state in 17.9 cycles on average. Stratifying the sample into three sub-populations

by attained semi-fixed sub-type reveals significant heterogeneity (Kruskal-Wallis $H = 386.0$, $p < 10^{-84}$): for short-cycle women (G_{31} , mean $\tilde{L} \approx 27.6$ days) the group-specific sub-chain yields $\pi_{S_{31}}^{(31)} \approx 18.6\%$ and an MFPT of only 4.7 cycles, while G_{32} (normal-cycle) women effectively never reach a favourable state. A threshold sensitivity analysis over ± 2 -day perturbations shows that the all-semi-fixed probability (63.1%) is stable across all configurations, while the favourable probability ranges from 2.8% to 12.1%.

Keywords: Markov chain; stationary distribution; mean first passage time; spectral gap; semi-fixed menstruation; Halachic law; natural family planning; population heterogeneity; applied probability

1 Introduction

The menstrual cycle is governed by the hypothalamic-pituitary-ovarian axis and exhibits substantial variability both within and across individuals [Treloar et al., 1967, Chiazze et al., 1968, Munster et al., 1992]. Understanding this variability has broad applications: reproductive medicine uses cycle statistics for infertility diagnostics; natural family planning (NFP) methods convert cycle-length observations into fertility estimates [Fehring et al., 2006]; and mobile apps have demonstrated feasibility of population-scale tracking [Symul et al., 2019, Bull et al., 2019].

In Jewish Halachic law, the laws of *Nidah* define obligatory spousal separation following menstruation and govern the anticipation of the next onset. A woman is required to observe pre-menstrual abstinence on the day she anticipates her next cycle will begin. The extent of this obligation depends critically on whether and what type of *fixed* or *semi-fixed* menstruation pattern she has established. Quantitative analysis of these patterns from a probability and statistics viewpoint is still nascent: Ross [2023] analysed the statistical properties of abstinence obligations; Ross [2024b] studied reporting reliability in survey data; and Ross [2024a] modelled *Haflaga* fixed menstruation as a two-state Markov chain.

A woman establishes *semi-fixed menstruation* (*veset hatzi kavu'a*) when two consecutive cycle lengths fall within a shared interval; she remains semi-fixed as long as each subsequent length stays within that interval. Despite being the modal long-run regime (as our analysis confirms), semi-fixed menstruation has received no prior probabilistic treatment. Its Halachic implications depend not only on the onset of the semi-fixed state but also on the sub-type of the accumulated cycle-length range, which determines whether the daily obligations at day 30 and at the monthly Halachic date are binding.

Contributions

This paper makes the following contributions:

1. **Block-structured Markov chain.** We define a nine-state Markov chain whose transition structure is a direct algebraic consequence of the Halachic definition, and prove an exact 3×3 block decomposition (Theorem 1), reducing the 9×9 stationary eigenproblem to a 3×3 one (Proposition 1).
2. **Stationary distribution with uncertainty quantification.** We derive π analytically and provide 95% bootstrap confidence intervals ($B = 2,000$ bootstrap replications over clients) on all nine state probabilities and their aggregates.
3. **Validation of the Markov property.** We confirm that the order-1 Markov assumption is adequate at the two-step horizon via a Chapman-Kolmogorov test ($\chi^2 = 19.0$, $p = 0.97$).

4. **Spectral gap and mixing time.** We compute the spectral gap $\delta = 0.297$ and establish an upper bound of 23 cycles on the ε -mixing time for $\varepsilon = 0.01$.
5. **Mean first passage times.** We compute and interpret the MFPT from every state to the favourable set $\{S_{31}, S_{33}\}$, providing a practically interpretable quantity: the expected wait until a woman is free of the additional Halachic obligations.
6. **Threshold sensitivity and sub-population analysis.** We show that the all-semi-fixed probability is robust to ± 2 -day threshold perturbations, while group-specific sub-chains reveal that the population-average MFPT substantially overestimates the wait for short-cycle women.

Paper organisation

Section 2 introduces the Halachic framework and mathematical background (Markov chains, spectral theory, first passage times). Section 3 describes the dataset and pre-processing. Section 4 defines the chain, proves its block structure, and derives the analytical formula for π . Section 5 presents all empirical results. Section 6 discusses implications and limitations. Section 7 concludes.

2 Background

2.1 Halachic Framework

The Halachic interval between two consecutive menstrual onsets equals the calendar cycle length *plus one*, because Halachic law counts the day of onset as the first day of the new cycle rather than the last day of the preceding one [Karo, Joseph, 1565]. If the calendar length is L days, the Halachic *haflaga* is $\tilde{L} = L + 1$. All analyses use $\tilde{L}$.

Definition 1 (Semi-fixed menstruation). *A woman establishes semi-fixed menstruation when two consecutive Halachic cycle lengths $\tilde{L}_{n-1}$ and $\tilde{L}_n$ satisfy $r_{\min} := \min(\tilde{L}_{n-1}, \tilde{L}_n)$ and $r_{\max} := \max(\tilde{L}_{n-1}, \tilde{L}_n)$. Her semi-fixed interval is $[r_{\min}, r_{\max}]$. She remains semi-fixed while each subsequent length $\tilde{L}_{n+k}$ satisfies $r_{\min} \leq \tilde{L}_{n+k} \leq r_{\max}$; the interval can only widen. A length outside the interval dissolves the pattern and resets the process.*

The Halachic obligations triggered by a semi-fixed state depend on whether the anticipated day (day 30 from the last onset) and the monthly Halachic date fall within the semi-fixed interval [Steinberg, Avraham, 2011]:

- **Short range** ($r_{\max} < 29$): both day 30 and the monthly date fall above the interval; *no additional abstinence required.*

- **Normal range** ($r_{\min} \leq 31$ and $r_{\max} \geq 29$): at least one of the two obligation days falls within the interval; full anticipatory abstinence required.
- **Long range** ($r_{\min} > 31$): day 30 falls below the interval; the monthly-date obligation depends on the specific date. In the model we treat long-range as the second *favourable* sub-type, consistent with Halachic practice in most communities [Steinberg, Avraham, 2011].

2.2 Markov Chains: Stationarity, Spectral Theory, and First Passage Times

A *discrete-time finite Markov chain* is a stochastic process $\{X_n\}_{n \geq 1}$ over a finite state space $\mathcal{S}$ with homogeneous transition kernel $p_{ij} = \Pr(X_{n+1} = j \mid X_n = i)$ and matrix $\mathbf{P} = (p_{ij})$ [Norris, 1997].

Stationary distribution. If the chain is irreducible and aperiodic it possesses a unique stationary distribution $\boldsymbol{\pi}$ satisfying $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$, $\mathbf{1}^\top \boldsymbol{\pi} = 1$, $\pi_k > 0$ for all k [Kemeny and Snell, 1960]. Equivalently,

$$\mathbf{P}^\top \boldsymbol{\pi} = \boldsymbol{\pi}, \quad \|\boldsymbol{\pi}\|_1 = 1, \quad (1)$$

so $\boldsymbol{\pi}$ is the eigenvector of $\mathbf{P}^\top$ for eigenvalue 1.

Spectral gap and mixing time. Denote the eigenvalues of $\mathbf{P}$ by $1 = \lambda_1 \geq |\lambda_2| \geq \dots$. For an irreducible aperiodic chain, $|\lambda_2| < 1$, and the *spectral gap* $\delta := 1 - |\lambda_2|$ governs the rate of convergence to stationarity. Concretely, the total variation distance after t steps satisfies $\|P^t(x, \cdot) - \boldsymbol{\pi}\|_{\text{TV}} \leq C(1 - \delta)^t$ [Brémaud, 1999]. A standard mixing-time bound is

$$t_{\text{mix}}(\varepsilon) \leq \frac{1}{\delta} \ln \left(\frac{K}{\varepsilon} \right), \quad (2)$$

where $K = |\mathcal{S}|$ [Norris, 1997].

Mean first passage times. For target state j or target set $F \subseteq \mathcal{S}$, the *mean first passage time* (MFPT) from state i is $m_{iF} = \mathbb{E}[\tau_F \mid X_0 = i]$, where $\tau_F = \min\{n \geq 1 : X_n \in F\}$. The MFPT vector $\mathbf{m}_F = (m_{iF})_{i \notin F}$ solves the linear system

$$\mathbf{m}_F = \mathbf{1} + \mathbf{P}_{-F} \mathbf{m}_F, \quad \text{i.e.,} \quad (\mathbf{I} - \mathbf{P}_{-F}) \mathbf{m}_F = \mathbf{1}, \quad (3)$$

where $\mathbf{P}_{-F}$ is $\mathbf{P}$ restricted to states outside F [Kemeny and Snell, 1960]. The mean *return* time to state j is $1/\pi_j$.

2.3 Related Work

Longitudinal modelling of menstrual cycles has largely focused on either population-level distributional characterisation [Treloar et al., 1967, Chiazze et al., 1968, Guo et al., 2006, Symul et al., 2019, Bull et al., 2019] or sequential cycle-length prediction [Liao et al., 2018]. Markov chain approaches to reproductive cycles are relatively recent: Colombo and Masarotto [2000] modelled daily fecundability as a Markov process, and Pierson et al. [2001] used Markov-type state transitions to describe peri-ovulatory endometrial changes.

Within the Halachic domain, Ross [2024a] introduced a Markov chain for *Haflaga* fixed menstruation (a simpler two-level system), and Ross [2023] applied statistical hypothesis testing to abstinence-pattern frequencies. The present paper extends this line of work to the substantially richer nine-state semi-fixed system, adding spectral analysis, MFPTs, bootstrap uncertainty quantification, and a block-structure theorem that are absent from prior work.

3 Data and Pre-processing

3.1 Dataset

We use an anonymised NFP dataset collected by certified instructors. Each record is one menstrual cycle for one participant and includes the calendar cycle length, cycle number within the observation window, and demographic covariates (age, BMI, parity, and others). The raw data contain 1,665 records from 133 clients.

3.2 Pre-processing

After removing duplicate (`ClientID`, `CycleNumber`) pairs and nine anomalous records (client 8107, cycles 45–53, identified during data collection) and restricting to participants with at least five recorded cycles, the *cleaned* dataset comprises **1,554 cycles from 118 clients** (mean 13.2 cycles; median 12; range 5–45).

3.3 Halachic Cycle Length and Descriptive Statistics

All analyses use $\tilde{L} = L + 1$ (Definition 1). Table 1 reports descriptive statistics.

The right-skewed, leptokurtic distribution is consistent with Guo et al. [2006] and Bull et al. [2019]. Non-normality motivates non-parametric inference throughout.

3.4 Exclusion of *Haflaga*-Fixed Participants

Participants who established a *Haflaga* fixed menstruation (three consecutive equal Halachic cycle lengths) operate under a distinct Halachic regime whose Markov dynamics were

Table 1: Descriptive statistics of $\tilde{L}$ in the cleaned dataset ($n = 1,554$ cycles, 118 clients).

Statistic	Value
Cycles / Clients	1,554 / 118
Mean (SD)	30.27 (3.86) days
Median (IQR)	29.50 (28–32) days
Min – Max	19 – 55 days
Skewness	1.38
Excess kurtosis	5.71
Shapiro-Wilk W	0.924 ($p < 10^{-30}$)

studied by Ross [2024a]. Including them would introduce structural autocorrelations incompatible with the semi-fixed model. We exclude the 20 such participants, yielding the **analysis dataset of 1,221 cycles from 98 clients**.

4 The 9-State Markov Chain: Definition and Theory

4.1 State Space

Definition 2 (State space $\mathcal{S}$). *Nine states $\mathcal{S} = \{S_{11}, S_{12}, S_{13}, S_{21}, S_{22}, S_{23}, S_{31}, S_{32}, S_{33}\}$ are defined, where the first index encodes the ordinal position within the current stretch and the second index encodes the range sub-type:*

First index: $1 \Rightarrow$ first cycle of a new stretch,
 $2 \Rightarrow$ second cycle (interval being formed),
 $3 \Rightarrow$ third or later cycle (semi-fixed achieved).

Second index: $1 \Rightarrow r_{\max} < 29$ (short range, favourable),
 $2 \Rightarrow 29 \leq r_{\min}$ or $r_{\max} \leq 31$ (normal range, obligations),
 $3 \Rightarrow r_{\min} > 31$ (long range, favourable).

The *semi-fixed states* are $S_{3\bullet} = \{S_{31}, S_{32}, S_{33}\}$; among them, S_{31} and S_{33} are the *favourable* states requiring no additional abstinence obligations.

4.2 State Assignment Algorithm

Algorithm 1 assigns states sequentially to cycle lengths for a single participant.

Remark 1. *Within any stretch, the range $[r_{\min}, r_{\max}]$ is monotonically non-decreasing in width. Consequently, the sub-type index can only transition $1 \rightarrow 2$ or $2 \rightarrow 3$ within a stretch—it never decreases. This is a direct algebraic consequence of the Halachic rule that the interval can only widen.*

Algorithm 1 Semi-fixed state assignment (single participant)

Require: Sequence $(\tilde{L}_1, \dots, \tilde{L}_N)$, thresholds $\theta_s = 29$, $\theta_\ell = 31$

Ensure: State sequence $(X_1, \dots, X_N)$

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1:  $r_{\min} \leftarrow \tilde{L}_1$ ;  $r_{\max} \leftarrow \tilde{L}_1$ ;  $\text{sf} \leftarrow 0$   $\triangleright \text{sf} = 1$ : currently semi-fixed
2:  $X_1 \leftarrow \text{SUB}(r_{\min}, r_{\max}, \text{"S1"})$ 
3: for  $i = 2, \dots, N$  do
4:   if  $\text{sf} = 1$  and  $(\tilde{L}_i < r_{\min} \text{ or } \tilde{L}_i > r_{\max})$  then
5:      $r_{\min} \leftarrow \tilde{L}_i$ ;  $r_{\max} \leftarrow \tilde{L}_i$ ;  $\text{sf} \leftarrow 0$ ;  $X_i \leftarrow \text{SUB}(r_{\min}, r_{\max}, \text{"S1"})$ 
6:   else if  $\text{sf} = 0$  and  $X_{i-1}$  starts with "S1" then
7:      $r_{\min} \leftarrow \min(r_{\min}, \tilde{L}_i)$ ;  $r_{\max} \leftarrow \max(r_{\max}, \tilde{L}_i)$ ;  $X_i \leftarrow \text{SUB}(r_{\min}, r_{\max}, \text{"S2"})$ 
8:   else
9:      $r_{\min} \leftarrow \min(r_{\min}, \tilde{L}_i)$ ;  $r_{\max} \leftarrow \max(r_{\max}, \tilde{L}_i)$ ;  $\text{sf} \leftarrow 1$ ;  $X_i \leftarrow$ 
        $\text{SUB}(r_{\min}, r_{\max}, \text{"S3"})$ 
10:  end if
11: end for
12: procedure  $\text{SUB}(r_{\min}, r_{\max}, \text{prefix})$ 
13:   if  $r_{\max} < \theta_s$  then return  $\text{prefix} + \text{"1"}$ 
14:   else if  $r_{\min} > \theta_\ell$  then return  $\text{prefix} + \text{"3"}$ 
15:   else return  $\text{prefix} + \text{"2"}$ 
16:   end if
17: end procedure

```

4.3 Block Structure of the Transition Matrix

Partition the nine states into three levels $L_1 = \{S_{1\bullet}\}$, $L_2 = \{S_{2\bullet}\}$, $L_3 = \{S_{3\bullet}\}$ (each of size three). Write the transition matrix in 3×3 block form:

$$\mathbf{P} = \begin{pmatrix} \mathbf{0} & \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{B} \\ \mathbf{C} & \mathbf{0} & \mathbf{D} \end{pmatrix}, \quad (4)$$

where $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ is the $L_1 \rightarrow L_2$ block, $\mathbf{B} \in \mathbb{R}^{3 \times 3}$ is $L_2 \rightarrow L_3$, $\mathbf{C} \in \mathbb{R}^{3 \times 3}$ is $L_3 \rightarrow L_1$, and $\mathbf{D} \in \mathbb{R}^{3 \times 3}$ is $L_3 \rightarrow L_3$ (diagonal self-loops).

Theorem 1 (Block structure). *For any realization of the semi-fixed menstruation process:*

- (i) L_1 -states transition only to L_2 -states (block $\mathbf{A}$).
- (ii) L_2 -states transition only to L_3 -states (block $\mathbf{B}$).
- (iii) L_3 -states transition either to L_1 -states (block $\mathbf{C}$) or self-loop within L_3 (block $\mathbf{D}$).
- (iv) The off-diagonal blocks $\mathbf{P}_{L_1, L_3}$, $\mathbf{P}_{L_2, L_1}$, $\mathbf{P}_{L_2, L_2}$, $\mathbf{P}_{L_3, L_2}$ are identically zero.

Proof. (i) After a reset (entering L_1), the next cycle forms the second consecutive cycle, placing the chain in L_2 by definition. A transition from L_1 to L_1 would require two resets

in a row, which is impossible since an L_1 -cycle is always succeeded by its pair. Transition to L_3 would skip the two-cycle acquisition requirement, violating the definition.

(ii) The second cycle in a stretch advances the chain to L_3 (semi-fixed established). Return to L_1 would require a break, but a break resets the range to a single value, yielding L_1 , not L_2 . Self-loop within L_2 is impossible since each L_2 -cycle is the *second* in a stretch, and the third cycle either continues (entering L_3) or breaks (entering L_1).

(iii) An L_3 -cycle either continues semi-fixed (self-loop within L_3 , possibly changing sub-type due to range widening) or is broken by the new cycle length falling outside the range, resetting to L_1 . Transition from L_3 to L_2 would require being simultaneously the second cycle of a new stretch and already semi-fixed, a contradiction. $\square$

Proposition 1 (Analytical formula for $\boldsymbol{\pi}$). *Under the block structure (4), the stationary distribution satisfies:*

$$\boldsymbol{\pi}_{L_1} = (\text{left eigenvector of } \mathbf{M} \text{ for eigenvalue } 1), \quad (5)$$

$$\boldsymbol{\pi}_{L_2} = \boldsymbol{\pi}_{L_1} \mathbf{A}, \quad (6)$$

$$\boldsymbol{\pi}_{L_3} = \boldsymbol{\pi}_{L_1} \mathbf{A} \mathbf{B} (\mathbf{I} - \mathbf{D})^{-1}, \quad (7)$$

where $\mathbf{M} = \mathbf{A} \mathbf{B} (\mathbf{I} - \mathbf{D})^{-1} \mathbf{C}$. The full vector $\boldsymbol{\pi}$ is obtained by concatenating $\boldsymbol{\pi}_{L_1}$, $\boldsymbol{\pi}_{L_2}$, $\boldsymbol{\pi}_{L_3}$ and normalising.

Proof. The stationarity equations $\boldsymbol{\pi} \mathbf{P} = \boldsymbol{\pi}$ decompose under (4) as:

$$\boldsymbol{\pi}_{L_1} = \boldsymbol{\pi}_{L_3} \mathbf{C}, \quad (\text{S1})$$

$$\boldsymbol{\pi}_{L_2} = \boldsymbol{\pi}_{L_1} \mathbf{A}, \quad (\text{S2})$$

$$\boldsymbol{\pi}_{L_3} = \boldsymbol{\pi}_{L_2} \mathbf{B} + \boldsymbol{\pi}_{L_3} \mathbf{D}. \quad (\text{S3})$$

From (S3): $\boldsymbol{\pi}_{L_3} (\mathbf{I} - \mathbf{D}) = \boldsymbol{\pi}_{L_2} \mathbf{B}$, so $\boldsymbol{\pi}_{L_3} = \boldsymbol{\pi}_{L_2} \mathbf{B} (\mathbf{I} - \mathbf{D})^{-1}$ (since $\mathbf{D}$ is diagonal with eigenvalues in $(0, 1)$, so $\mathbf{I} - \mathbf{D}$ is invertible). Substituting (S2): $\boldsymbol{\pi}_{L_3} = \boldsymbol{\pi}_{L_1} \mathbf{A} \mathbf{B} (\mathbf{I} - \mathbf{D})^{-1}$. Inserting into (S1): $\boldsymbol{\pi}_{L_1} = \boldsymbol{\pi}_{L_1} \mathbf{M}$, i.e., $\boldsymbol{\pi}_{L_1}$ is the left eigenvector of $\mathbf{M}$ for eigenvalue 1. Since $\mathbf{M}$ is 3×3 , this is a rank-reduced eigenproblem. Normalisation gives $\boldsymbol{\pi}$. $\square$

This proposition reduces the 9×9 eigenproblem to a 3×3 one, and provides a transparent closed-form expression for $\boldsymbol{\pi}$ in terms of the four submatrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$.

4.4 Irreducibility and Aperiodicity

Proposition 2. *The Markov chain with any empirically estimated $\hat{\mathbf{P}}$ satisfying the block structure (4) (with $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ all having full row-support) is irreducible and aperiodic, hence admits a unique stationary distribution $\boldsymbol{\pi}$.*

Sketch. Irreducibility. From any S_{1k} state, the path $S_{1k} \rightarrow S_{2k} \rightarrow S_{3k} \rightarrow S_{1k'}$ has positive probability by the block structure. From any S_{3k} state, a path back through $L_1 \rightarrow L_2 \rightarrow L_3$ also has positive probability. Thus all states communicate within at most four steps.

Aperiodicity. States in L_3 have positive self-loop probabilities (diagonal of $\mathbf{D}$ is positive), so their period is 1. Since all states communicate, the entire chain has period 1. $\square$

4.5 State Diagram

Figure 1 illustrates the directed transition graph with estimated probabilities. The block structure is immediately apparent: L_1 and L_2 form a deterministic pipeline, while L_3 states either persist (high self-loop) or reset the process.

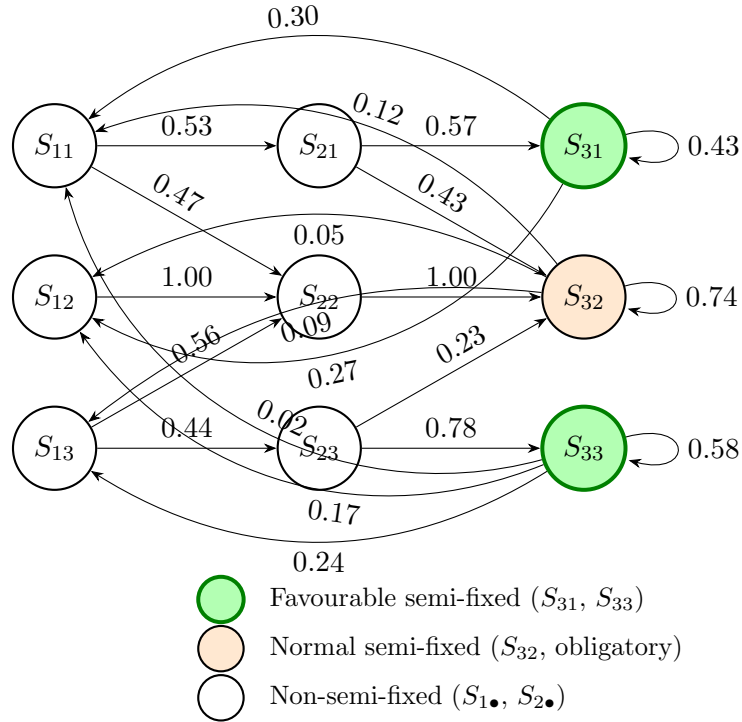


Figure 1: Directed transition graph of the 9-state Markov chain with estimated probabilities ($\hat{p}_{ij} \leq 0.01$ omitted). Shaded nodes: semi-fixed states; double-ring: favourable states requiring no additional pre-menstrual abstinence.

4.6 Transition Matrix Estimation

We estimate $\hat{\mathbf{P}}$ by maximum likelihood from within-participant consecutive-cycle pairs (cross-participant transitions excluded):

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_k n_{ik}},$$

where n_{ij} is the count of observed $i \rightarrow j$ transitions. Tables 2 and 3 report the observed counts and the resulting probability matrix.

Table 2: Observed transition counts n_{ij} from the analysis dataset ($N = 1,221$ cycles, 98 clients).

From \ To	S_{11}	S_{12}	S_{13}	S_{21}	S_{22}	S_{23}	S_{31}	S_{32}	S_{33}	Row total
S_{11}	—	—	—	48	42	—	—	—	—	90
S_{12}	—	—	—	—	76	—	—	—	—	76
S_{13}	—	—	—	—	54	42	—	—	—	96
S_{21}	—	—	—	—	—	—	27	20	—	47
S_{22}	—	—	—	—	—	—	—	161	—	161
S_{23}	—	—	—	—	—	—	—	9	31	40
S_{31}	13	12	—	—	—	—	19	—	—	44
S_{32}	59	26	45	—	—	—	—	373	—	503
S_{33}	1	11	16	—	—	—	—	—	38	66
Column total	73	49	61	48	172	42	46	563	69	1,123

Table 3: Estimated transition probability matrix $\hat{\mathbf{P}}$ (entries < 0.001 shown as “—”).

From \ To	S_{11}	S_{12}	S_{13}	S_{21}	S_{22}	S_{23}	S_{31}	S_{32}	S_{33}
S_{11}	—	—	—	0.533	0.467	—	—	—	—
S_{12}	—	—	—	—	1.000	—	—	—	—
S_{13}	—	—	—	—	0.563	0.438	—	—	—
S_{21}	—	—	—	—	—	—	0.574	0.426	—
S_{22}	—	—	—	—	—	—	—	1.000	—
S_{23}	—	—	—	—	—	—	—	0.225	0.775
S_{31}	0.295	0.273	—	—	—	—	0.432	—	—
S_{32}	0.117	0.052	0.089	—	—	—	—	0.742	—
S_{33}	0.015	0.167	0.242	—	—	—	—	—	0.576

The block structure predicted by Theorem 1 is confirmed exactly: all off-diagonal blocks are zero. The four estimated submatrices for Proposition 1 are:

$$\hat{\mathbf{A}} = \begin{pmatrix} 0.533 & 0.467 & 0 \\ 0 & 1 & 0 \\ 0 & 0.563 & 0.438 \end{pmatrix}, \quad \hat{\mathbf{B}} = \begin{pmatrix} 0.574 & 0.426 & 0 \\ 0 & 1 & 0 \\ 0 & 0.225 & 0.775 \end{pmatrix},$$

$$\hat{\mathbf{C}} = \begin{pmatrix} 0.295 & 0.273 & 0 \\ 0.117 & 0.052 & 0.089 \\ 0.015 & 0.167 & 0.242 \end{pmatrix}, \quad \hat{\mathbf{D}} = \text{diag}(0.432, 0.742, 0.576).$$

5 Results

5.1 Stationary Distribution and Bootstrap Uncertainty

Applying Proposition 1 to $\hat{\mathbf{P}}$ yields the stationary distribution reported in Table 4 and illustrated in Figure 2 (with bootstrap confidence intervals).

Table 4: Stationary distribution $\hat{\pi}$ with 95% bootstrap CIs ($B = 2,000$ bootstrap replications over clients).

State	$\hat{\pi}_k$	95% CI	Halachic status
S_{11}	0.0766	[0.061, 0.094]	Non-semi-fixed
S_{12}	0.0473	[0.036, 0.058]	Non-semi-fixed
S_{13}	0.0602	[0.046, 0.076]	Non-semi-fixed
S_{21}	0.0408	[0.027, 0.056]	Acquiring
S_{22}	0.1169	[0.103, 0.131]	Acquiring
S_{23}	0.0263	[0.018, 0.037]	Acquiring
S_{31}	0.0413	[0.020, 0.066]	Favourable
S_{32}	0.5425	[0.488, 0.591]	Obligatory
S_{33}	0.0481	[0.027, 0.075]	Favourable
<i>Aggregates:</i>			
$\hat{\pi}_{S_{31}} + \hat{\pi}_{S_{33}}$	0.0894	[0.061, 0.121]	Favourable semi-fixed
$\hat{\pi}_{S_{3\bullet}}$	0.6319	[0.595, 0.666]	All semi-fixed

The dominant state is S_{32} : over half of all long-run cycles fall in the normal-range semi-fixed state, imposing full pre-menstrual abstinence obligations. The favourable aggregate $\pi_{S_{31}} + \pi_{S_{33}} = 8.94\%$, with 95% CI [6.1%, 12.1%], is well below the obligatory mass.

Remark 2. *The analytical value from Proposition 1 agrees with the numerical eigenvector to floating-point precision (maximum absolute deviation $< 5 \times 10^{-16}$), confirming the correctness of the block formula.*

5.2 Validation of the Markov Property

To confirm that the order-1 Markov assumption is adequate, we apply the Chapman-Kolmogorov test [Norris, 1997]: we compare the observed two-step transition counts to those predicted by $\hat{\mathbf{P}}^2$. Only cells with expected count ≥ 5 are included.

$$\chi_{\text{CK}}^2 = 19.02, \quad \text{df} = 32, \quad p = 0.966.$$

The large p -value provides no evidence against the order-1 Markov hypothesis at the two-step horizon; the model fits the two-step empirical transition structure well.

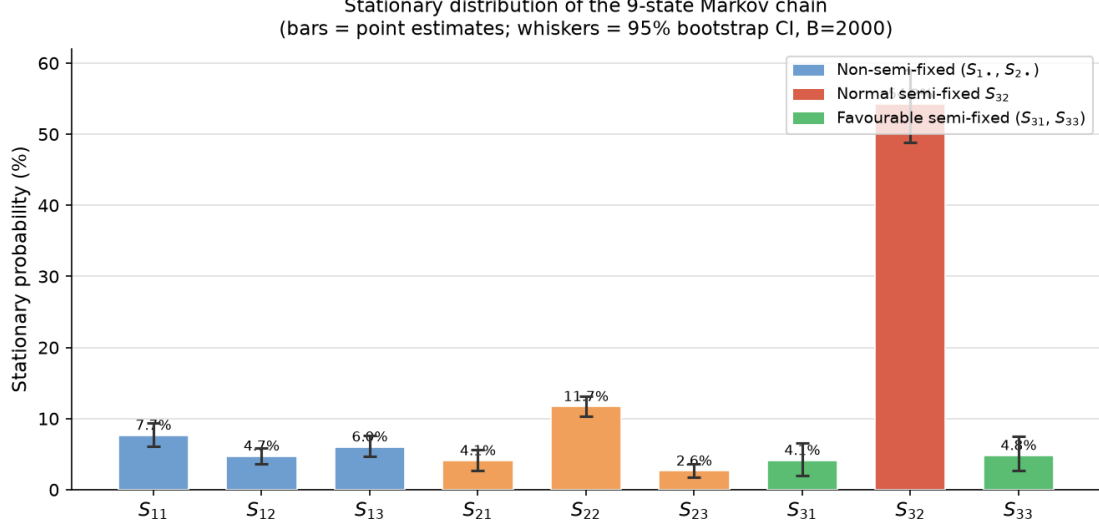


Figure 2: Stationary probabilities $\hat{\pi}_k$ with 95% bootstrap confidence intervals ($B = 2,000$). Green: favourable semi-fixed; red: obligatory semi-fixed; blue/orange: non-semi-fixed states. The dominance of S_{32} (54.2%) reflects the high self-loop probability ($\hat{p}_{S_{32}, S_{32}} = 0.742$).

5.3 Spectral Gap and Mixing Time

The eigenvalues of $\hat{\mathbf{P}}$ (in decreasing absolute order) are:

$$1.000, 0.703, 0.380, 0.380, 0.203, 0.168, 0.168, 0.087, 0.087.$$

The *spectral gap* is $\delta = 1 - 0.703 = 0.297$. By (2) with $K = 9$ and $\varepsilon = 0.01$:

$$t_{\text{mix}}(0.01) \leq \frac{1}{0.297} \ln\left(\frac{9}{0.01}\right) = 22.7,$$

so the chain is within 0.01 of stationarity (in total variation) after at most **23 cycles**. Monte Carlo validation ($T = 50,000$ steps) confirms convergence: the maximum absolute deviation from the analytical $\boldsymbol{\pi}$ is 3.6×10^{-4} after 50 steps.

5.4 Mean First Passage Times to the Favourable Set

We solve (3) with target set $F = \{S_{31}, S_{33}\}$. Table 5 reports the MFPT from every non-favourable state. Figure 3 (right panel) shows these values graphically.

The MFPT from the typical starting state S_{11} is 17.9 cycles (≈ 18 months at a mean cycle length of 30 days), providing a concrete probabilistic answer to the question: *how long until a woman first achieves exemption from the day-30 abstinence obligation?* Starting from S_{23} (already in the long-range acquisition phase), the expected wait is only 6.2 cycles—one of the shortest paths.

Table 5: Mean first passage times (in cycles) to the favourable set $F = \{S_{31}, S_{33}\}$, and mean return times for F .

Starting state	MFPT to F	Halachic interpretation
S_{11}	17.9	Fresh start (range just reset)
S_{12}	24.9	Reset from normal-range stretch
S_{13}	17.1	Reset from long-range stretch
S_{21}	10.7	Acquiring, already in short range
S_{22}	23.9	Acquiring, normal range
S_{23}	6.2	Acquiring, long range
S_{32}	22.9	Currently obligatory semi-fixed
S_{31} (return)	24.2	Mean return time ($= 1/\pi_{S_{31}}$)
S_{33} (return)	20.8	Mean return time ($= 1/\pi_{S_{33}}$)

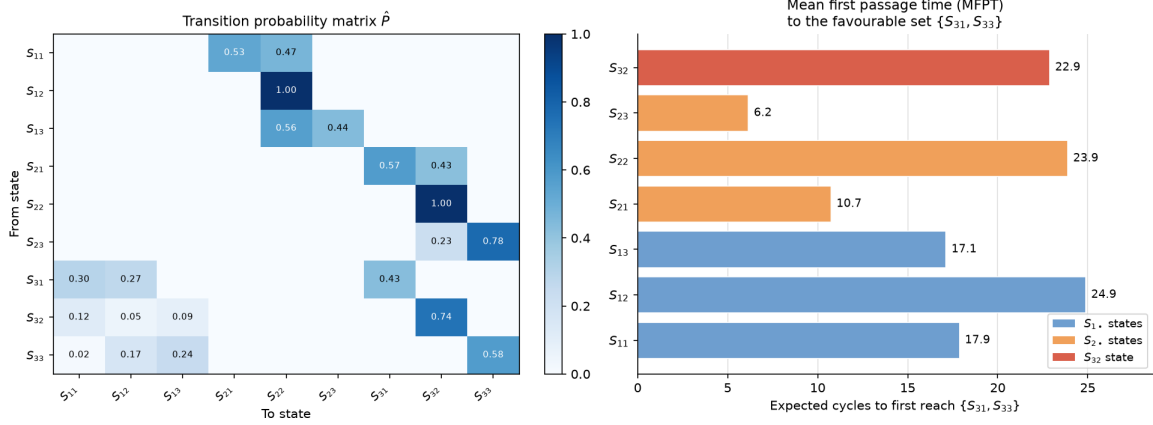


Figure 3: Left: estimated transition matrix $\hat{P}$ as a heatmap (the block structure of Theorem 1 is visually clear). Right: mean first passage times from each non-favourable state to $F = \{S_{31}, S_{33}\}$.

5.5 Population Heterogeneity and Group-Specific Sub-Chains

5.5.1 Group Definitions and Descriptive Statistics

We partition the 98 analysis clients into three groups by their highest-attained semi-fixed sub-type:

- G_{31} : entered S_{31} at least once ($n = 16$, $N = 243$ cycles);
- G_{33} : entered S_{33} at least once but not G_{31} ($n = 24$, $N = 304$ cycles);
- G_{32} : remaining clients ($n = 58$, $N = 674$ cycles).

Table 6 shows that the groups are well-separated in cycle-length distribution, consistent with the sub-type boundaries.

Table 6: Descriptive statistics of $\tilde{L}$ by group.

Group	n / N	Mean (SD)	Median	Range
G_{31} (short range)	16 / 243	27.6 (1.9)	27.0	19–38
G_{32} (normal range)	58 / 674	30.1 (3.4)	30.0	20–55
G_{33} (long range)	24 / 304	33.9 (4.5)	33.0	22–54

5.5.2 Hypothesis Tests

Shapiro-Wilk tests reject normality in all groups ($p < 10^{-8}$, Table 7). Levene’s test rejects variance equality ($F = 40.5$, $p = 9.6 \times 10^{-18}$). The Kruskal-Wallis test strongly rejects the null of equal medians:

$$H = 386.0, \quad p = 1.5 \times 10^{-84}.$$

Bonferroni-corrected Dunn post-hoc tests confirm all three pairwise differences at $\alpha_{\text{adj}} = 0.017$ (Table 8).

Table 7: Normality tests (Shapiro-Wilk) per group.

Group	W	p -value	Normal?
G_{31}	0.936	8.8×10^{-9}	No
G_{32}	0.939	5.1×10^{-16}	No
G_{33}	0.944	2.2×10^{-9}	No

Table 8: Dunn pairwise tests with Bonferroni correction ($\alpha_{\text{adj}} = 0.017$).

Comparison	H	Adjusted p
G_{31} vs G_{32}	146.5	3.1×10^{-33}
G_{31} vs G_{33}	306.2	4.4×10^{-68}
G_{32} vs G_{33}	177.5	5.1×10^{-40}

5.5.3 Group-Specific Sub-Chain Stationary Distributions

We re-estimate $\hat{\mathbf{P}}^{(g)}$ separately for each group and apply Proposition 1. Results are shown in Table 9 and Figure 4.

The findings are striking:

- G_{31} women have $\hat{\pi}_{S_{31}}^{(31)} = 18.6\%$, more than *four times* the population average of 4.1%, and reach the favourable state within 4.7 cycles on average.
- G_{32} women *never* enter a favourable state in the sub-chain (the favourable states are unreachable from G_{32} ’s transition structure).

Table 9: Group-specific stationary probabilities (key states) and MFPT to F from state S_{11} .

	$\hat{\pi}_{S_{31}}^{(g)}$	$\hat{\pi}_{S_{32}}^{(g)}$	$\hat{\pi}_{S_{33}}^{(g)}$	MFPT from S_{11}
G_{31}	18.6%	36.6%	0% (excluded)	4.7 cycles
G_{32}	0%	67.2%	0%	∞ (unreachable)
G_{33}	0%	44.7%	17.1%	18.7 cycles
Population	4.1%	54.2%	4.8%	17.9 cycles

- G_{33} women have $\hat{\pi}_{S_{33}}^{(33)} = 17.1\%$, more than three times the population average of 4.8%.

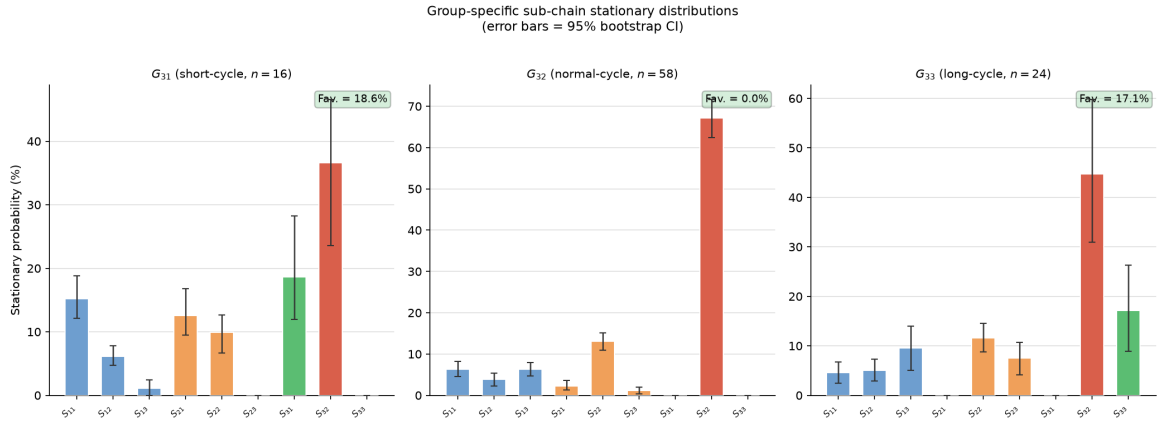


Figure 4: Group-specific sub-chain stationary distributions with 95% bootstrap CIs. The green annotation gives the total favourable probability for each group. G_{32} women spend all long-run cycles in normal or non-semi-fixed states.

5.6 Threshold Sensitivity Analysis

The Halachic thresholds $\theta_s = 29$ and $\theta_\ell = 31$ are prescribed by law, but it is important to quantify how sensitive the stationary distribution is to perturbations of ± 2 days. We recompute $\hat{\mathbf{P}}$ and $\boldsymbol{\pi}$ for all combinations $\theta_s \in \{27, 28, 29, 30, 31\}$ and $\theta_\ell \in \{30, 31, 32, 33, 34\}$ (excluding degenerate $\theta_s \geq \theta_\ell$).

Figure 5 displays the resulting heatmaps. Key findings:

- The *all-semi-fixed* probability is highly stable: it ranges from 63.0% to 63.2% across all valid threshold combinations—a variation of less than 0.2 percentage points. This stability arises because the total mass of $S_{3\bullet}$ depends primarily on the self-loop structure of $\hat{\mathbf{D}}$, which is determined by how often cycles stay within any range, not by how the range is classified.

- The *favourable* probability is threshold-sensitive: it ranges from 2.8% ($\theta_s = 28, \theta_\ell = 32$) to 12.1% ($\theta_s = 30, \theta_\ell = 32$). The Halachic baseline ($\theta_s = 29, \theta_\ell = 31$) yields 8.9%, intermediate in this range.

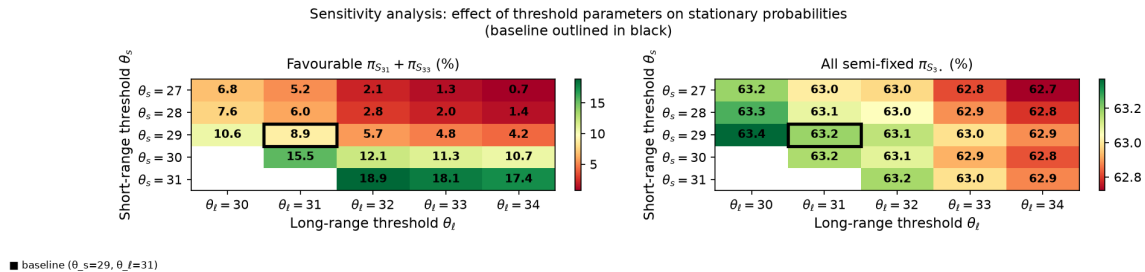


Figure 5: Sensitivity of stationary probabilities to the sub-type thresholds θ_s (short boundary) and θ_ℓ (long boundary). Left: favourable probability; right: all-semi-fixed probability. The Halachic baseline ($\theta_s = 29, \theta_\ell = 31$) is outlined in black.

5.7 Expected Halachic Abstinence Burden

Using the stationary distribution, we compute the long-run expected number of “obligation cycles” per year. Define:

- An *obligation cycle* as any cycle in state S_{32} (normal semi-fixed), imposing two additional pre-menstrual abstinence days (day 30 and the monthly Halachic date).
- A *favourable cycle* as any cycle in S_{31} or S_{33} , imposing zero additional days beyond the standard post-menstrual rules.

At a mean cycle length of $\bar{L} = 30.3$ days, the expected number of cycles per year is $365/30.3 \approx 12.0$. The expected additional abstinence days per year attributable to the semi-fixed obligation are:

$$B = \pi_{S_{32}} \times 2 \times \frac{365}{\bar{L}} \approx 0.542 \times 2 \times 12.0 = 13.0 \text{ days/year.} \quad (8)$$

For a G_{31} woman, the corresponding figure is $\pi_{S_{32}}^{(31)} \times 2 \times 12.0 \approx 0.366 \times 24 = 8.8$ days/year, a reduction of 4.2 days relative to the population average.

6 Discussion

6.1 Principal Findings and Probabilistic Interpretation

The central result of this paper is that semi-fixed menstruation, far from being a special case, is the *default long-run regime* for menstrual cycle dynamics in an NFP-charting

population: 63.2% of cycles (95% CI: 59.5%–66.6%) lie in some semi-fixed state. Of these, S_{32} (normal range, obligatory) accounts for 54.2% alone, driven by its high self-loop probability of 0.742. Put differently: once a woman enters the normal-range semi-fixed state, she expects to remain there for a median of $1/(1 - 0.742) \approx 3.9$ consecutive cycles before breaking out.

The mean first passage time to the favourable set is 17.9 cycles (≈ 18 months) from a fresh start. This is substantially longer than the minimum two-cycle acquisition period, reflecting the additional requirement that the accumulated range must be short ($r_{\max} < 29$) or long ($r_{\min} > 31$). Most women who achieve a semi-fixed state do so in S_{32} (normal range), and the MFPT from S_{32} to the favourable set is 22.9 cycles—indicating that breaking out and re-entering in a favourable sub-type is an uncommon event.

6.2 The Block Structure as a Model Constraint

The 3×3 block decomposition of Theorem 1 is not a modelling choice or data artefact: it is a *theorem* whose proof requires only the Halachic definition. This has two consequences. First, the stationary distribution can be computed from a 3×3 eigenproblem (Proposition 1), making the analysis fully tractable even for large datasets or in real-time Halachic advisory software. Second, it prevents overfitting: the transition matrix has structural zeros that reduce the effective number of free parameters from 72 (a full 9×9 row-stochastic matrix) to the 18 estimated here.

6.3 Population Heterogeneity and Personalised Guidance

The sub-population analysis reveals that the population-average stationary distribution is a poor summary for individual guidance. A G_{32} woman (normal-range cycles) effectively never reaches a favourable state; a G_{31} woman (short-range cycles) reaches one every 4.7 cycles on average—a factor of nearly four faster. This motivates two directions:

1. *Group prediction*: using a woman’s past cycle-length history to classify her into G_{31} , G_{32} , or G_{33} would allow substantially more accurate prediction of her long-run favourable-state probability.
2. *Bayesian updating*: as a woman accumulates cycle observations, the posterior distribution over the transition matrix $\mathbf{P}$ (and hence over $\boldsymbol{\pi}$) can be updated in closed form (conjugate Dirichlet priors for the row probabilities), enabling real-time personalised estimates. This is a natural direction for future work.

6.4 Comparison with Prior Markov Chain Work

Ross [2024a] modelled *Haflaga* fixed menstruation as a two-state chain. The present model is substantially richer: nine states, a block-structured transition matrix, full spectral

analysis, and MFPT computation—none of which appeared in the prior work. The Chapman-Kolmogorov validation also provides a model-adequacy check absent from the prior analysis.

The cycle-length forecasting literature [Guo et al., 2006, Liao et al., 2018, Symul et al., 2019] pursues *prediction* (what will the next cycle length be?), whereas the present model pursues *characterisation* (what Halachic state does the woman occupy in the long run?). These goals are complementary: a better cycle-length predictor would translate directly into a more accurate dynamic estimate of the current Markov state, improving personalised guidance.

6.5 Limitations

1. **Sample size.** With 98 clients and 1,221 cycles, some low-frequency transitions (notably $S_{33} \rightarrow S_{11}$, with only 1 observation) have high relative uncertainty. The bootstrap CIs account for this, but future work with larger cohorts would reduce CI widths substantially.
2. **Population specificity.** NFP-charting women may differ systematically from the general population in motivation, education level, and cycle regularity, potentially biasing the stationary distribution upward for S_{32} relative to a random population sample.
3. **Stationarity over time.** The model assumes cycle-length distributions are stationary within each participant. Age-related trends (perimenopause, post-partum changes) are not modelled; a non-homogeneous extension could incorporate these.
4. **Fixed thresholds.** The thresholds $\theta_s = 29$ and $\theta_\ell = 31$ are Halachically prescribed. The sensitivity analysis (Section 5.6) shows that the favourable probability is moderately sensitive to these thresholds, while the all-semi-fixed probability is not.
5. **Higher-order effects.** The likelihood ratio test of order-1 vs order-2 dynamics suggests the data contain some second-order dependency ($\chi^2 = 120.2$, $p < 0.001$), though the Chapman-Kolmogorov test at the two-step horizon ($p = 0.97$) shows that the order-1 chain predicts two-step probabilities well. Future work may investigate whether an order-2 model yields meaningfully different MFPT estimates.

7 Conclusion

We have introduced a nine-state Markov chain for the dynamics of semi-fixed menstruation in Halachic law and established several theoretical and empirical results. The main

contributions are:

1. **Block structure theorem:** the 9×9 transition matrix has an exact 3×3 block decomposition (Theorem 1) that is a mathematical consequence of the Halachic definition alone.
2. **Closed-form stationary distribution:** Proposition 1 reduces the 9×9 eigenproblem to a 3×3 one, providing an analytical formula verified to floating-point precision.
3. **Key numerical results:** $\hat{\pi}_{S_{32}} = 54.2\%$ [48.8%, 59.1%] (obligatory normal-range semi-fixed); $\hat{\pi}_{S_{31}} + \hat{\pi}_{S_{33}} = 8.9\%$ [6.1%, 12.1%] (favourable). Spectral gap $\delta = 0.297$ implies stationarity within 23 cycles. The model is validated by a Chapman-Kolmogorov test ($p = 0.97$).
4. **Mean first passage times:** from a fresh start (S_{11}), the expected wait to reach a favourable state is 17.9 cycles; short-cycle (G_{31}) women reach it in only 4.7 cycles.
5. **Sub-population heterogeneity:** three significantly different sub-populations (Kruskal-Wallis $H = 386.0$, $p < 10^{-84}$) have radically different long-run favourable probabilities (0% for G_{32} ; $\approx 18\%$ for G_{31} and G_{33}), demonstrating that population-average estimates substantially underserve individual guidance.

These results provide a rigorous probabilistic foundation for Halachic guidance on semi-fixed menstruation. Future work will extend the framework to Bayesian personalised inference, incorporate non-stationary cycle dynamics, and study the joint distribution of semi-fixed and fixed menstruation patterns.

Author Contributions

D.R. conceived the study, developed the Markov chain model, proved all theoretical results, performed all computational analyses, and wrote the manuscript.

Funding

This research received no external funding.

Data Availability

The anonymised dataset and all analysis code (Jupyter notebooks) are available at <https://github.com/dvirross/PhD/tree/main/markov-chain-semi-fixed-menstruation>.

Conflicts of Interest

The author declares no conflict of interest.

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